

RetroDiff: Deriving the Retrieval-Augmented Training Objective

We derive the training objective for **RetroDiff**, a retrieval-augmented diffusion model. Let x denote a clean data sample, x_t its noised counterpart at step t , q the forward noising distribution, p_θ the learned reverse distribution with parameters θ , and T the total number of diffusion steps.

Step 1: Variational lower bound. Because the marginal likelihood $p_\theta(x)$ is intractable, we lower-bound it via Jensen’s inequality applied to the expectation over the forward process $q(x_{1:T} \mid x)$:

$$\log p_\theta(x) \geq \mathbb{E}_{q(x_{1:T}|x)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x)} \right] = -\mathcal{L}_{\text{ELBO}} \quad (1)$$

Maximising $\log p_\theta(x)$ is therefore equivalent to minimising $\mathcal{L}_{\text{ELBO}}$.

Step 2: Decomposition into tractable terms. Expanding the joint $p_\theta(x_{0:T}) = p(x_T) \prod_t p_\theta(x_{t-1} \mid x_t)$ and the forward posteriors $q(x_{t-1} \mid x_t, x)$, which are Gaussian in closed form, yields the canonical decomposition:

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_q[-\log p_\theta(x_0 \mid x_1)]}_{\text{reconstruction}} + \sum_{t=2}^T \underbrace{\text{KL}(q(x_{t-1} \mid x_t, x) \parallel p_\theta(x_{t-1} \mid x_t))}_{\text{step-}t \text{ denoising}} \quad (2)$$

The reconstruction term trains p_θ to recover x_0 from a single noisy observation x_1 ; each KL term penalises deviation of the learned reverse step from the tractable forward posterior.

Step 3: Retrieval-augmented objective (RetroDiff). Equations (1)–(2) treat samples in isolation. RetroDiff’s key contribution is to condition the per-sample ELBO on a set $R(x)$ of k retrieved exemplars drawn from the training corpus. Averaging over the data distribution gives the final objective:

$$\mathcal{L}_{\text{RetroDiff}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\mathcal{L}_{\text{ELBO}}(x \mid R(x))] \quad (3)$$

By conditioning on $R(x)$, the model gains access to high-quality denoising hints at every diffusion step, reducing the effective complexity of each KL term in (2) without modifying the model architecture.